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3. (a) **Table 3.1** shows the classification of **three** species of badger.

Table 3.1

	Species of badger		
Classification	European	Honey	Asian
Kingdom	Animal	Animal	Animal
Phylum	Vertebrate	Vertebrate	Vertebrate
Class	Mammal	Mammal	Mammal
Order	Carnivore	Carnivore	Carnivore
Family	Mustelidae	Mustelidae	Mustelidae
Genus	Meles	Mellivora	Meles
Species	meles	capensis	leucurus

(i) State the **two** species of badger that are most closely related.
Explain your answer. [2]

....

....

....

....

(ii) Use **Table 3.1** to state the scientific name of the honey badger. [1]

....

(iii) State **one** reason why scientists use scientific names instead of common names for organisms. [1]

....

....

(iv) The honey badger is a vertebrate. State what is meant by the term vertebrate. [1]

....

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(b) **Image 3.2** is a fact file about the honey badger.

Image 3.2

Common names: honey badger, ratel, honey ratel

Body length: approximately 70 cm (males bigger than females)
Habitat: varies from desert to rainforest
Diet includes: insect larvae, scorpions, lizards, rodents, birds, snakes, foxes and wild cats
Predators include: lion, leopard, humans

Other facts:

- Their skin is very tough. This makes it hard for snakes to bite.
- They live on their own most of the time. This reduces intraspecific competition.
- They hunt for their own food but they will also steal prey from other carnivores.
- Their sharp teeth and long claws allow them to easily rip meat from bone.

Use the information in **Image 3.2** to describe **two** adaptations of the honey badger. Explain how each helps them survive. [2]

Adaptation	How it helps them to survive
....	....
....	....

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(c) Suggest what would happen to the number of honey badgers if the number of lions increased. Explain your answer. [2]

....

....

....

....

(d) (i) State what is meant by **intraspecific competition**. [1]

....

....

(ii) List **three** resources for which **all** animals compete. [3]

1. ....
2. ....
3. ....

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